Order Request Management

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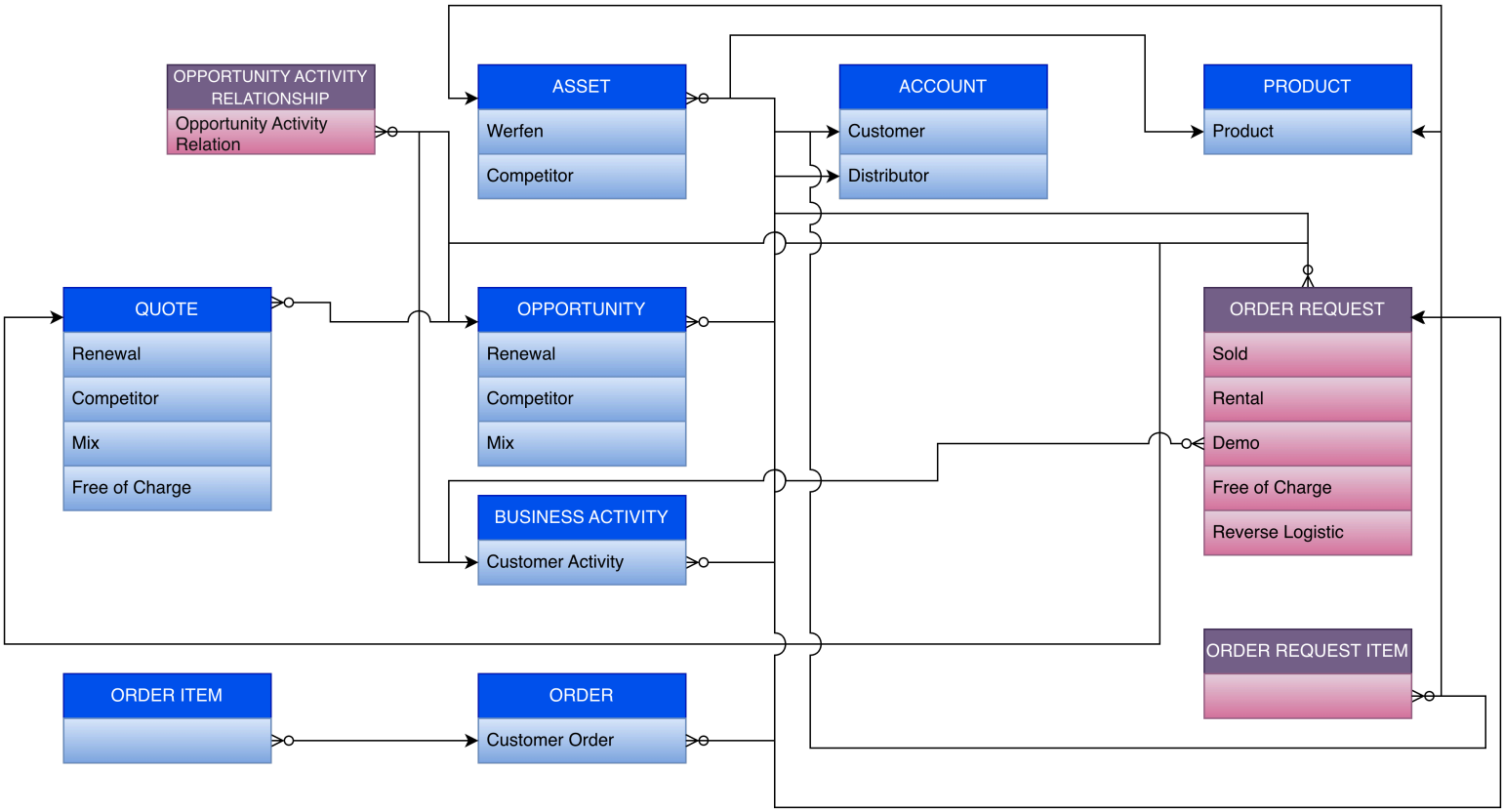

Data Model

- **Account**: Customer or Distributor for whom the order request is opened.
- **Opportunity**: Open Opportunity upon which request is made.
- **Quote**: Quote configured for the customer/distributor.
- **Opportunity Activity Relationship**: It allows you to link an opportunity with an activity.
- **Business Activity**: Planned activities of an account.
- **Asset**: Account instruments
- **Product**: Type of products
- **Order Request**: new custom object to record order request
- **Order Request Item**: links the assets or products related to the order request and account address (ship to).
- **Order**: Orders validated and generated in the SAP system.
- **Order Items**: links the products related to the order and generated in the SAP system.

Order Request Types

There are different types of order requests that can be made:

- Sold (Instruments)
- Rental (Instruments)
- Demo (Instruments)
- Free of Charge (Without instruments, they use a demo for the instrument)
- Reverse Logistic (return)

The origin of a request varies depending on the type:

- **Account**: Sold, Rental, Demostration, Free of Charge, Reverse Logistic
- **Quote**: Sold, Rental, Free of Charge, Demostration. ***You can only generate requests for accepted offers.***
- **Opportunity**: Sold, Rental, Demostration, Free of Charge
- **Business Activity** (Demostration): Demostration

Mandatory Relationship & Validation Matrix

The Order Request object will enforce the following **mandatory relationships** based on the **Request Type**.

1. Sold & Rental

These types require a direct link to a successful negotiation.

- **Mandatory Account**
- **Open Opportunity**
- **Awarded Quote:**
  - The linked **Quote** must have its status set to **Awarded**.
  - **Exception Handling:** If the Quote is *not* Awarded, the user must populate the **Explanation** (Long Text) field.
  - **UI Guidance:** A dynamic help message or validation error will display: *"The quote is not awarded yet. Please, correct or explain the reason why you're requesting the instrument."*

2. Demonstration (Demo)

This involves a **customer demostration**, which will **always require an instrument request**.

- **Mandatory Account**
- **Business Activity:** Must link to a specific record of type **Demonstration**.
- **Open Opportunity:** The request must be justified by an Opportunity related to the aforementioned Demonstration Activity.

3. Free of Charge (FOC)

- **Mandatory Account**
- **Open Opportunity**
- **Hardware Restriction:**
  - **Validation Rule:** The **Order Request Items** cannot contain **Instruments**.
  - **Logic:** If the user attempts to add an instrument to an FOC request, the system will block the save and suggest: *"Instruments must be requested via a 'Demonstration' Order Request type."*

4. Reverse Logistics

Simplified for returns and pickups.

- **Mandatory Account:** The only required relationship is the **Account** from which the material is being collected.

Sold & Rental

This refers to a instrument **sales or instrument rental request.**

The process can be initiated from the customer account, an open opportunity, or a customer quote. In cases where the request originates from an opportunity or quote, relationships with said record (such as related products) and all possible fields will be populated automatically.

Additionally, it **is mandatory to associate an Quote in "Awarded" status**.

Demostration

This involves a **customer demostration**, which will **always require an instrument request**.

It is also mandatory to associate a demostration-type activity related to the open opportunity for which the demo is required; therefore, the origin of a demostration can be initiated from the customer record page or from a demostration activity. If a demostration activity is not available, one must be generated to proceed with completing the record information.

Free of Charge

These are **free-of-charge orders where no instrument is required**. If an instrument is required, a Demo request must be performed instead. While it is not necessary to associate a demostration activity, the request must be

linked to an open customer opportunity.

More Information: Free of Charge Process Introduction NoA December 2024.pptx

Reverse Logistic

This refers to an **instrument return**. The request originates from the customer requiring the return, allowing for the selection of different instruments installed at the customer site (which are located in the **Asset** object, related to the customer account).

Order Request Fields

To ensure the seamless management of different order request types, the custom object **Order Request** shall be provisioned with the following fields. These fields are designed to capture transactional data and enforce business logic based on the origin of the request.

Field Label	Data Type	Logic / Functional Behavior
Request Type	Picklist	**Controlled by RecordType** (Values: *Sold, Rental, Demostration, Free of Charge, Reverse Logistics*).
Request Date	Date	Defaults to **System.Today()** upon record creation.
Status	Picklist	Lifecycle stages: *Draft, Under Approval, Send to SAP, Ordered, Shipped, Delivered, Rejected*.
Quote	Lookup (Quote)	Auto-populated if the request originates from an Accepted Quote.
Opportunity	Lookup (Opportunity)	Auto-populated if initiated from an Opportunity or related Quote.
Account	Lookup (Account)	Inherited from the origin record (Account, Opportunity, Quote, or Activity).
Ship to	Lookup (Address)	Customer addresses available for the account
Werfen Address	Checkbox	This indicates that one of the Werfen addresses available for the region/country will be used. Therefore, it is not necessary to fill in the Ship TO field, and this will be provided by the customer master in SAP.
Business Unit	Global Picklist	Auto-populated if the request originates from an Accepted Quote or Opportunity.
Region/Country	Text (Formula)	Automatically derived from the associated Account's location data.
Care Of	Text	Designated recipient (Person or Department). May include Werfen employees.
Customer Order	Text	Free text for the Customer's Purchase Order (PO) number *(specific to 'Sold' types)*.
Preferred Delivery Date	Date	Requested date for fulfillment.
Comments	Text Area	Additional transactional notes.
Explanation	Text Area	Comments to justify a Sold or Rental order request that must be made before the Quote status is "Awarded"

Approval Fields

To support the **Dynamic Approval Matrix Management** architecture, the **Order Request** object requires a set of technical fields. These fields are populated automatically by a Before-Save Flow during record creation or specific updates, serving as the static routing criteria for the Standard Approval Process.

Field Label	Data Type	Logic / Functional Behavior
Approval Matrix	Lookup (Approval Matrix)	**Auto-populated.** Direct link to the configuration record used to define the routing. Provides an audit trail for which rules were applied.
Threshold L2	Currency (16,2)	**Auto-populated.** The minimum transaction amount required to trigger the 2nd level of approval.
Threshold L3	Currency (16,2)	**Auto-populated.** The minimum transaction amount required to trigger the 3rd level of approval.
Approver L1	Lookup (Group)	**Auto-populated.** Base level approver (Queue).
Approver L2	Lookup (Group)	**Auto-populated.** Escalation approver for Level 2.
Approver L3	Lookup (Group)	**Auto-populated.** Final escalation approver for Level 3.

UI & Security Logic

- **Field-Level Security (FLS):** These fields act as the technical engine for the approval process and should be set to **Read-Only** for all standard users (Sales, Supply Chain). Only the System Administrator profile and the internal Flow automation should have edit access to prevent manual tampering.
- **Page Layout:** It is recommended to group these fields in a dedicated, collapsible section at the bottom of the Order Request page layout (e.g., *System Information - Approval Routing*) to keep the commercial interface clean while allowing auditors or managers to review the routing logic.
- **Technical Tip (Salesforce Lookups to Queues):** Salesforce does not natively support custom Lookup fields pointing directly to the `Group` (Queue) object. The standard workaround for `Approver L1/L2/L3` is to either use a standard User lookup (if assigning individuals) or, to support Queues, use **Text (18)** fields to store the Queue IDs. The Standard Approval Process can then reference these text fields to route the step to the dynamically assigned Queue.

More details: [Dynamic Approval Matrix Management]

Werfen IT & Connectivity Section

These fields follow specific **Field Level Dependency** and **Visibility Rules**:

- **Connection to Werfen IT Solution:** Checkbox.
- **Werfen IT Solution:** Multi-select Picklist (*HemoHub, GemWebPlus, GemWebLive, QuantaLink, ImmuLink, MatchX, Modulab*).
  - *Requirement: Only editable/visible if "Connection to Werfen IT Solution" is TRUE.*
- **Connectivity to Competitive IT Solution:** Checkbox.
- **Competitor:** Text.
  - *Requirement: Only editable/visible if "Connectivity to Competitive IT Solution" is TRUE.*
- **Competitive IT Solution:** Text.
  - *Requirement: Only editable/visible if "Connectivity to Competitive IT Solution" is TRUE.*
- **Werfen Address:** Checkbox.
- **Ship to:** Lookup.
  - *Requirement: Only editable/visible if "Werfen Address" is FALSE.*

For the Werfen IT Solution and Competitive IT fields, we will use Dynamic Forms in Lightning App Builder. This will allow the text fields and multipicklist to remain hidden until the user selects the corresponding checkbox, reducing visual clutter in the layout.

Special Requirements for FOC & Demo Requests

To ensure proper auditing and alignment with business goals, **Free of Charge (FOC)** and **Demostration (Demo)** order request types will require specific data points that differ from standard "Sold" or "Rental" transactions.

Field Label	Data Type	Requirement / Logic
Account Key	Picklist	**Mandatory for FOC & Demo.** Values: *Sample, Sales Agreement, Product Failure, Evaluations/Studies, Validations/Verifications, Internal Consumption*.
Reason	Text Area	Detailed justification for the non-revenue request.
Activity	Lookup (Business Activity)	**Mandatory for "Demo" type only.** Must reference a record of type "Demostration". This field remains hidden for "Free of Charge" requests.

Dynamic Forms (UI Visibility):

- The **Account Key** and **Reason** fields will only appear on the Page Layout when the **Request Type** is "Free of Charge" or "Demostration".
- The **Activity** lookup will be dynamically rendered *only* when the **Request Type** is "Demostration".

Validation Rules (Data Integrity):

- **Demo-Activity Constraint:** A rule will prevent the submission of a "Demostration" request if the **Activity** field is null or if the linked Activity record is not of the "Demostration" Type.
- **FOC-Account Key Requirement:** A rule will ensure **Account Key** is populated whenever **Request Type** is **Free of Charge** or **Demostration**.

Special Requirements for Reverse Logistic

The **Reverse Logistics** request type is designed to manage the return, replacement, or repair of equipment and materials. This configuration requires specific categorization while streamlining the interface to remove unnecessary commercial data.

Field Label	Data Type	Logic / Functional Behavior
Request Reason	Picklist	**Mandatory for Reverse Logistics.** Values: *Lost, Replacement, Upgrade, Renewal, Reparation*.
Preferred Pick Up Date	Date	Request collection date.

To ensure a clean user experience and data integrity, the following visibility rules will be enforced via **Dynamic Forms** or **Record Types**:

- **Conditional Field Activation:** The **Request Reason** field will be dynamically displayed and marked as required only when the **Request Type** is set to **Reverse Logistics**.
- **Field Exclusion:** To prevent confusion during the return process, the following commercial lookup fields will be **hidden** for this request type:
  - **Opportunity**
  - **Quote**

Summary of Request Type Constraints

Request Type	Opportunity / Quote	Activity (Demo)	Account Key	Request Reason	Preffered Delivery Date	Preffered Pick Up Date
Sold	Visible	Hidden	Hidden	Hidden	Visible	Hidden
Rental	Visible	Hidden	Hidden	Hidden	Visible	Hidden
Demonstration	Visible	**Mandatory**	**Mandatory**	Hidden	Visible	Hidden
Free of Charge	Visible	Hidden	**Mandatory**	Hidden	Visible	Hidden
Reverse Logistics	**Hidden**	Hidden	Hidden	**Mandatory**	**Hidden**	**Visible**

The `ExternalId` field must be populated using a **Before-Save Record-Triggered Flow** upon record creation. By concatenating the Order Request ID and the User ID, the database engine will automatically block any attempt to subscribe the same user to the same order twice, throwing a standard DML duplicate error.

Order Request Item Fields

The **Order Request Item** custom object will capture the granular details of the products included in a request. This object maintains a child relationship with the **Order Request** (Header) and serves as the bridge to the **Product** master data.

Field Label	Data Type	Logic / Functional Behavior
Order Request	Master-Detail (Order Request)	Order Request related
Product	Lookup (Product2)	Selection of the specific product to be requested.
Material	Formula (Text)	Retrieves `Product.ProductCode` (SAP Material Master code).
Description	Formula (Text)	Retrieves `Product.Description` (SAP Material Master description).
Quantity	Number (18, 0)	The specific quantity of the unit requested.
Unit Price	Currency (16, 2)	Inherited from the Quote; **Mandatory** if Request Type is "Sold".
Total Price	Formula (Currency)	`Quantity * UnitPrice`
Preferred Delivery Date	Date	Defaults to the Header's date but allows for line-level overrides.
Ship To	Lookup (WERFEN_obj_address__c)	Defaults to the Header's address but allows for line-level overrides.
Werfen Address	Checkbox	This indicates that one of the Werfen addresses available for the region/country will be used. Therefore, it is not necessary to fill in the Ship TO field, and this will be provided by the customer master in SAP.
Business Unit	Global Picklist	Auto-populated if the request originates from an Accepted Quote or Opportunity.
Preferred Serial Number	Text	Optional field to specify a specific hardware unit.
Placement Type	Picklist	(Values: *Renewal, New/Incremental, Competitive*).
Competitor	Text	Indicate the competitor you are competing against for the instrument. Reporting is only provided for placement type = competitive

Specialized Item Requirements (FOC, Demo & Product Failure)

To support quality control and specific logistical needs for non-revenue orders, the **Order Request Item** object will include conditional fields and streamlined layouts based on the **Account Key**.

Field Label	Data Type	Requirement / Logic
Desired Lot	Number (18, 0)	**Optional.** Specific to **FOC** and **Demo** types. Allows the user to request a specific batch for evaluations or validations.
Pick-up Site	Lookup (WERFEN_obj_address__c)	**Specific to "Product Failure" logic.** Technically uses the same **Ship_To__c** field but relabeled via Dynamic Forms to reflect the return/collection nature of the request. Reporting is only provided for placement type = competitive

"Product Failure" Streamlined Logic

When the **Account Key** is set to **Product Failure**, the business requirement is to simplify the item entry to focus solely on the faulty material and its collection. In this scenario, the Item UI will be restricted to the following subset of fields:

1. **Product, Material and Description:** Identification of the failed unit.
2. **Desired Lot:** Critical for quality tracking to identify the specific batch that failed.
3. **Quantity:** Number of units to be replaced or returned.
4. **Pick-up Site:**
    - **Automation:** Defaults to the **Ship To** address defined in the **Order Request** header.
    - **Flexibility:** Can be manually modified if the collection point differs from the delivery point.

Specialized Item Requirements (Reverse Logistic)

The **Order Request Item** interface for "Reverse Logistics" will be strictly limited to tracking physical assets and coordinating their return. This layout focuses on hardware identification rather than financial pricing.

Field Label	Data Type	Logic / Functional Behavior
Order Request	Master-Detail (Order Request)	Order Request related
Product	Lookup (Product2)	Identification of the product model to be returned. Autopopulate from Asset
Material	Formula (Text)	Retrieves `Product.ProductCode`.
Description	Formula (Text)	Retrieves `Product.Description`.
Asset	Lookup (Asset)	**Mandatory.** Filtered to show Assets associated with the Header Account.
Instrument Status	Formula (Text)	Retrieves `Asset.WESAL_fld_instrumentStatus__c`
Serial Number	Formula (Text)	Retrieves `Asset.SerialNumber` to ensure the correct unit is tracked.
Preferred Pick Up Date	Date	Inherited from Header but editable per line.
Ship To	Lookup (Address__c)	Inherited from Header; defines the collection point. Editable per line.
Status	Picklist	Lifecycle stages: *Draft, Under Approval, Send to SAP, Ordered, Shipped, Delivered, Rejected*.

Order Request Item Lifecycle Tracking

To manage the 1:N relationship between the **Order Request** (Salesforce) and the **Orders** (SAP), the **Order Request Item** object will act as the primary tracking entity. This ensures that if part of a request is delayed or shipped separately, the status is accurately reflected.

Field Specification: Line-Item Status

- **Item Status:** Picklist.
  - **Values:** *Draft, Pending Approval, Approved, Sent to SAP, SAP Confirmed, Shipped, Delivered, Cancelled*.

Synchronization Logic: The Split-Order Flow

Because the distribution happens at the moment of transmission, the status updates will follow this flow:

1. **Approval:** When the **Order Request** (Header) is approved, all **Order Request Items** transition to *Ordered*.
2. **Splitting & Transmission:** The integration service groups items by _Ship-to.
3. **Fulfillment:** When SAP synchronizes the resulting **Orders** back to Salesforce, a trigger will match the items (using the **Order Request** lookup) and update the **Order Request Item Status** to **Shipped** or **Delivered** (depend by Order Status).

UI/UX Logic: Simplified Item Grid

When the parent record is a **Reverse Logistics** request, the entry grid will be stripped of commercial fields to reduce data entry errors. The following fields are **excluded** from this view:

- **Quantity**
- **Unit Price**

- **Total Price**
- **BackOrder**

Order Entry Component (Grid Interface)

The interface for managing **Order Request Items** shall be a dynamic, data-table-style component with the following capabilities:

- **Bulk Actions:** Options to **Select All** and **Deselect All** for batch processing.
- **Row Management:** Buttons to manually **Add Rows** or **Delete Rows**.
- **Data Sourcing:**
  - If no **Quote** is provided in the header, the table remains empty for manual entry.
  - If a **Quote** is linked, the table pre-populates with the corresponding quote lines.
- **Granular Logistics:**
  - **Per-Line Delivery Date:** Supports staggered delivery schedules within a single order.
  - **Per-Line Shipping Destination:** Enables split shipments (e.g., some units to the Client, others to Werfen Facilities).

Transactional Actions

The UI will feature two primary action triggers:

1. **Save as Draft:** Persists the current data to Salesforce records in **Status = Draft**.
2. **Submit for Approval:** Validates the data and initiates the specialized approval engine.

Multi-Tiered Approval Engine

Approvals will be managed natively within Salesforce using a flexible architecture that routes requests based on the following criteria:

- **Affiliate:** Geographic/Legal entity routing.
- **Request Type:** Different workflows for *Sold* vs. *Rental* vs. *Demo*.
- **Line of Business (LOB):** Routing to specific Business Unit managers.

Approver Workspace:

A centralized **Approval Dashboard** will be implemented within the Salesforce menu, listing all "Pending Approval" requests. This provides a unified view for managers to review and act upon requests without navigating to individual records. It will be done using the standard **"Items to Approve"** component on the Home Page, but customized with a **Lightning Page** that filters by "Order Request" object to ensure managers only see relevant commercial approvals in that specific menu.

SAP Integration (Final Step)

The **Final Approval Level** acts as the integration gateway to **SAP Supply Chain**:

- Upon final approval in Salesforce, the record is locked.
- An outbound integration triggers to SAP, transmitting the order request.
- SAP completes the remaining logistics fields and generates the official **Sales Order**.

SAP Integration & Order Partitioning Logic

To accommodate complex logistics where a single customer order involves multiple delivery destinations, the system must support distinct **Ship To** accounts/address at the line-item level.

To align with **SAP Sales Order** structures, which typically require a single "Sold To" per transaction, the integration shall execute a **Dynamic Splitting Logic** during transmission.

Integration Workflow & Execution

- **Line-Item Attribution:** Each **Order Request Item** will be associated with a specific destination address/account.
- **Logic of Partitioning:** At the moment of final approval and submission to SAP, the integration middleware (or Apex Integration Service) will group the items by their unique **Address.Account (Ship To)** identifier.
- **Transmission Output:** A single **Order Request** header in Salesforce will trigger multiple separate API calls or IDocs to SAP, resulting in multiple Sales Order numbers.
- **Reference Tracking:** The Salesforce Order Request record will maintain a related list or log of all generated SAP Sales Order IDs for full traceability.

Order Object (Inbound Data): Customer Order

Once an order is validated and processed in SAP, it will be synchronized back to Salesforce. This object represents the confirmed transactional record. The record type "Customer Order" will be assigned.

Field Label	Data Type	Logic / Functional Behavior
Account	Lookup (Account)	The customer account associated with the order.
Order Start Date	Standard Date	The initiation date of the order in SAP.
Order End Date	Standard Date	The expected or actual completion date.
SAP Order Number	Text (Unique, Ext. ID)	The primary key from SAP to prevent duplicates.
Business Unit	Global Picklist	Aligned with the Salesforce BU structure.
Status	Picklist	Values: *Pending/Partial, Shipped, Completed*.
Type	Picklist	Values: *Sold, Rental, Demo, Free of Charge, Reverse Logistics*.
Order Request	Lookup (Order Request)	Link to the original Salesforce request for traceability.

Order Item (Line Level)

The **Order Item** object provides granular visibility into fulfillment, including shipping details and inventory tracking (Lots/Delivery Notes).

Field Label	Data Type	Logic / Functional Behavior
Order	Lookup (Order)	Parent order reference.
Ship To	Lookup (WERFEN_obj_address__c)	The specific destination for this line item.
Werfen Address	Checkbox	This indicates that one of the Werfen addresses available for the region/country will be used. Therefore, it is not necessary to fill in the Ship TO field, and this will be provided by the customer master in SAP.
Business Unit	Global Picklist	Auto-populated if the request originates from an Accepted Quote or Opportunity.
Start Date	Date	Specific start date for the line item.
End Date	Date	Specific end date for the line item.
Delivery Note	Number (16, 2)	SAP Delivery Note reference number.
Lot	Number (18, 0)	Batch or Lot number for product traceability.
Product	Lookup (Product2)	Linked Salesforce Product record.
Material	Formula (Text)	Retrieves `Product.ProductCode`.
Description	Formula (Text)	Retrieves `Product.Description`.
Quantity	Number (18, 0)	Fulfilled quantity.
Unit Price	Currency (16, 2)	Final price as validated in SAP.
Total Price	Formula (Currency)	`Quantity * UnitPrice`.
Back Order	Checkbox	Indicator if the item is currently out of stock in SAP.

Additional Information

- More Information: 06 SF CORE TEMPLATE_Objects & Fields__Supply Chain Execution and Reverse Logistics v1.0.xlsx
- More Information: 06 SF CORE TEMPLATE_Objects & Fields__Supply Chain Execution and Reverse Logistics v2.0.xlsx